

ROUTE

Data Analysis Diploma Graduation Project Collection Analysis in Power BI

Business case, data guide, dashboard requirements and validation rules

Developed with the contribution of Route instructors and mentors

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The Business Case

A real estate company sells units using installment plans. The company needs one Power BI report to follow sales, issued installments, collected cash, unpaid amounts, bank exposure and project progress.

Your Mission

Build a complete collection dashboard from the supplied project sheets. Management should understand what became due, what was collected, what is still outstanding, and where the collection risk is.

Important

The reference pages show the expected business output. They are not DAX solutions. You must create the data preparation, model, measures and report logic yourself.

What success looks like

- A clean model that supports all projects with consistent logic.
- Correct treatment of paid, due and not-yet-due installments.
- Fast navigation between global pages and project detail pages.
- Results that reconcile with the control totals in the source data.

What does Collection mean?

Collection is the process of receiving money that customers are required to pay. Each sold unit has an installment schedule. When an installment becomes due, the company needs to know whether it was paid or is still outstanding.

Invoiced / Due

Value that has become payable according to the installment status.

Collected

Money that has already been received from customers.

Outstanding

Money that became due but has not been collected yet.

Collection Rate

How much of the issued invoices were successfully collected.

$$\text{Issued Amount} = \text{Collected Amount} + \text{Outstanding Amount}$$

3. Source Data Structure

Same 21 columns in every project sheet

Each project is stored in a separate worksheet. The worksheets follow the same structure, so the reporting logic should be reusable across projects.

Customer Name	Customer record; control rows are not customers.	Installment 1 to 7	Seven cells containing installment state and/or value.
Customer ID	Customer identification value.	Total Due Installments	Total amount currently due for the customer.
Unit Number	Business unit number.	Total Paid Installments	Total amount already collected.
Unit Code	Additional unit reference.	Bank	Cash or financing bank.
Unit Price	Selling price of the unit.	Sales Contract Date	Date of the sales contract.
Engineering Completion %	Completion value received from engineering.	POC	Percentage of completion used in the report KPI.
Payment Schedule	Installment percentage plan.	Sales Contract Number	Contract reference.
		Date	Reporting date used in date analysis.

4. The Three Control Rows

These rows are not customers

At the end of every project sheet, three special rows summarize the installment columns. Use them as control totals and exclude them from customer counts.

Summary Total

Total issued invoice amount for each installment.

Paid Installments

Total collected amount for each installment.

Remaining Installments

Total outstanding amount for each installment.

Required reconciliation

For every installment: $\text{Summary Total} = \text{Paid Installments} + \text{Remaining Installments}$.

5. Reading an Installment Cell

Classify the business state first

The installment columns are not simple numeric fields. A cell can represent three business states. Your logic must identify the state before calculating counts or amounts.

PAID

The installment is due and has been paid. It counts as issued and collected.

DUE - NOT PAID

The installment is due but has not been paid. It counts as issued and outstanding.

NOT DUE

The installment has not become payable yet. It must not count as an issued invoice.

Hint

Build one reliable classification rule and reuse it. Always exclude the three control rows from customer-level logic.

Users should move through the report with buttons, not by searching through Power BI page tabs.

2 Global Pages

Summary | Collection By Date Summary

4 Pages for Each Project

Overall | Bank | Cash | Bank-wise

8 projects x 4 detail pages = 32 project pages.

Core analytical pages: 34. Add a Home / navigation page for a total of 35 pages.

- Home button on every analytical page.
- Easy movement between the four pages of the same project.
- Consistent page titles, KPI order and visual structure.
- Keep project navigation simple and obvious.

7. Global Summary Page

One row per project

Outstanding balance	Cash	Bank	Amount collected	Invoiced Amount	Unit price	Sold%	#Sold	#Units	Project Name
38.52M	12.47M	26.05M	512.63M	548.17M	921.40M	88%	724	824	One Kattameya Compound
120.37M	34.28M	86.09M	276.45M	310.89M	532.15M	78%	491	631	Degla Landmark
22.18M	6.91M	15.27M	201.74M	225.36M	478.81M	72%	368	510	Lake Front 6 October
15.76M	7.02M	8.74M	648.37M	692.81M	1.03bn	90%	882	980	Rihana Avenue
26.94M	9.33M	17.61M	183.52M	206.77M	569.98M	63%	276	437	Skyline Katameya Compound
9.63M	3.29M	6.34M	115.87M	124.30M	395.17M	69%	223	325	Degla Palms 6 October Compound
11.42M	4.64M	6.78M	98.36M	107.49M	472.36M	81%	192	237	Zahra North Coast
8.13M	2.97M	5.16M	79.44M	86.12M	611.65M	76%	158	207	Crystal Plaza Maadi Compound
253.95M	80.91M	173.04M	2.12bn	2.30bn	5.02bn	79%	3,314	4,151	Total

What every column means

- # Units: fixed adopted total units, not a row count.
- Sold %: $\# \text{ Sold} / \text{fixed project units}$.
- Invoiced Amount: issued value across installments 1-7.
- Bank / Cash: outstanding split by payment method.
- # Sold: real customer rows only.
- Unit Price: total value of sold units.
- Amount Collected: money already received.
- Outstanding Balance: total unpaid due amount.

8. Collection By Date Summary

Old vs New based on the selected period

Collection By Date Summary			Cutoff Date : 7/1/2026 8/3/2026	
Project Name	Amount collected Old	Amount collected New	Outstanding balance Old	Outstanding balance New
One Kattameya Compound	512.67M	2.45M	28.94M	1.82M
Degla Landmark	873.34M	4.16M	45.78M	2.63M
Lake Front 6 October	1.024bn	8.73M	67.52M	6.18M
Rihana Avenue	618.90M	5.27M	31.84M	3.59M
Skyline Katameya Compound	396.28M	1.86M	22.17M	1.05M
Degla Palms 6 October Compound	270.59M	3.24M	15.62M	2.11M
Zahra North Coast	1.108bn	4.68M	48.95M	2.74M
Crystal Plaza Maadi Compound	49.37M	0.98M	3.44M	0.57M
Total	4.853bn	31.37M	264.26M	20.69M

OLD Records with Date before the selected Start Date.

NEW Records from Start Date through End Date, inclusive.

Required outputs

Amount Collected Old / New

Outstanding Balance Old / New

Hint: apply the date logic to real customer rows only. Do not include the three control rows in this analysis.

9. Project Overall Page

Installment performance for one project



Completion : 54.72%

No of units: 738

One Kattameya Compound

Total	7	6	5	4	3	2	1	Payments
	100%	85%	70%	60%	40%	20%	0%	Cumulative Due
	5%	5%	15%	15%	20%	20%	20%	Payments %
————	0	10	54	98	125	142	160	Issued Invoices
————	0	8	45	76	90	95	99	Collected Invoices
	0.00	1.20M	7.31M	11.84M	15.76M	16.83M	17.49M	Collected Amount
————	0%	80%	83%	78%	72%	67%	62%	Collection Rate
————	0	2	9	22	35	47	61	Outstanding Invoices
————	0.00	0.30M	1.49M	2.88M	6.01M	8.27M	10.82M	Outstanding Amount
————	0%	20%	17%	22%	28%	33%	38%	Outstanding %
————	0.00	1.50M	8.80M	14.72M	21.77M	25.10M	28.31M	Total Invoice Amount

Top KPIs

Project Completion % = average POC. # Units = fixed adopted project total.

For every installment 1 to 7

Issued Invoices | Collected Invoices | Collected Amount | Collection Rate | Outstanding Invoices | Outstanding Amount | Outstanding % | Total Invoice Amount

Hint: use installment state for invoice counts, and use the control totals to validate amount calculations.

10. Project Cash Page

Same KPIs, Cash segment only

 Completion : 57.84%		Degla Landmark (Cash)						No of units: 628	Degla Landmark
Total	7	6	5	4	3	2	1	Payments	
	100%	85%	70%	60%	40%	20%	0%	Cumulative Due	
	5%	5%	15%	15%	20%	20%	20%	Payments %	
	0	12	58	126	188	188	188	Issued Invoices	
	0	10	41	92	147	153	188	Collected Invoices	
	0.00	0.58M	6.17M	14.58M	24.54M	27.41M	33.69M	Collected Amount	
	0%	83%	71%	73%	78%	81%	100%	Collection Rate	
	0	2	17	34	41	35	0	Outstanding Invoices	
	0.00	0.12M	2.51M	5.41M	6.94M	4.38M	0.00	Outstanding Amount	
	0%	17%	29%	27%	22%	19%	0%	Outstanding %	
	0.00	0.70M	8.68M	19.99M	31.48M	31.79M	33.69M	Total Invoice Amount	

Requirement

Show the same installment KPIs as the Overall page, but only for customer records where the payment method is Cash.

Business reason

Management needs to compare direct-paying customers with bank-financed customers without changing the KPI definitions.

Hint: keep the business logic the same. Change only the segment/filter.

11. Project Bank Page

Same KPIs, financed customers only

 Completion : 54.27% Rihana Avenue (Bank) No of units: 678 Rihana Avenue								
Total	7	6	5	4	3	2	1	Payments
	100%	85%	70%	60%	40%	20%	0%	Cumulative Due
	5%	5%	15%	15%	20%	20%	20%	Payments %
	0	0	98	256	344	520	678	Issued Invoices
	0	0	72	214	297	467	599	Collected Invoices
	0.00	0.00	15.82M	63.47M	98.26M	143.94M	175.68M	Collected Amount
	0%	0%	73%	84%	86%	90%	88%	Collection Rate
	0	0	26	42	47	53	79	Outstanding Invoices
	0.00	0.00	5.91M	12.28M	16.20M	15.83M	21.84M	Outstanding Amount
	0%	0%	27%	16%	14%	10%	12%	Outstanding %
	0.00	0.00	21.73M	75.75M	114.46M	159.77M	197.52M	Total Invoice Amount

Requirement

Show the same installment KPIs as the Overall page for bank-financed customer records. Exclude Cash and blank bank values.

Business reason

This page shows the collection performance of financed sales and allows a direct comparison with Cash and Overall results.

Hint: the KPI definition must stay consistent across Overall, Cash and Bank pages.

12. Bank-wise Page

Analyze value and outstanding balance by bank

Bank Name	Unit Price	Invoiced Amount	Amount collected	Outstanding balance
البنك الأهلي المصري	7.85M	62.48M	60.27M	2.21M
بنك مصر	6.72M	48.63M	46.12M	2.51M
البنك العربي الأفريقي	10.15M	74.38M	71.63M	2.75M
بنك القاهرة	9.28M	53.16M	49.89M	3.27M
بنك QNB	8.67M	41.97M	40.20M	1.77M
كريدي اجريكول	7.35M	36.84M	35.02M	1.82M
EG Bank	7.10M	31.59M	29.74M	1.85M
CIB	12.40M	105.88M	101.12M	4.76M
بنك فيصل الإسلامي	5.05M	22.86M	21.71M	1.15M
Total	74.57M	477.79M	456.70M	21.09M

Required columns

Bank Name | Unit Price | Invoiced Amount | Amount Collected | Outstanding Balance

Business logic

Group customer records by bank. For each bank, calculate sold-unit value, issued installment value, money collected and remaining outstanding balance.

Validation hint: Invoiced Amount should reconcile with Amount Collected + Outstanding Balance according to the project data rules.

Issued Invoices

Count installments that are already due, whether paid or unpaid.

Collected Invoices

Count issued installments identified as paid.

Collected Amount

Value already received for the installment.

Collection Rate

Collected invoice count divided by issued invoice count.

Outstanding Invoices

Issued invoices that are still not collected.

Outstanding Amount

Issued amount that is still unpaid.

Outstanding %

Outstanding invoice count divided by issued invoice count.

Total Invoice Amount

Total issued amount for the installment.

Project Completion %

Average of the POC field for the project.

Sold %

Sold count divided by the fixed adopted project units.

Validation checklist

- Exclude the three control rows from customer counts.
- For each installment: Issued Amount = Collected Amount + Outstanding Amount.
- Outstanding Invoices = Issued Invoices - Collected Invoices.
- Cash Outstanding + Bank Outstanding = Total Outstanding.
- OLD and NEW follow the selected Start and End dates exactly.
- Sold % uses the fixed adopted unit count.
- Project Completion % uses average POC.
- Overall, Cash and Bank pages use the same KPI definitions.

Submission

Submit the final Power BI file with complete navigation, clean formatting, all required pages and results that pass the checks above

Goal: a manager should understand every project's collection position without opening the Excel source.